Skills Summary

Postulates, Theorems, Converses, and Biconditionals

Use this reference while completing your worksheet or when studying.

Skill 1 — Postulate or theorem, and using it

Postulate: accepted without proof — where the system starts (two points determine a line; the Segment and Angle Addition Postulates; the Linear Pair Postulate).

Theorem: proved from postulates, definitions and earlier theorems (vertical angles are congruent; supplements of the same angle are congruent).

The test: could someone write a proof of it? Then it is a theorem. Either way, both are legal reasons in a proof — the difference is where they came from.

Addition postulates: $AB + BC = AC$ when B is between A and C; $m\angle ABD + m\angle DBC = m\angle ABC$ when $\overrightarrow{BD}$ is interior.

Example: B is between A and C with $AB = 3x + 2$, $BC = 2x - 4$, and $AC = 23$. Find x.

Step 1. Betweenness is exactly the hypothesis of the Segment Addition Postulate, so the postulate converts the picture into the single equation $AB + BC = AC$.

Step 2. $(3x + 2) + (2x - 4) = 23$, so $5x - 2 = 23$ and $5x = 25$.

$\Rightarrow$ $\boldsymbol{x = 5}$

Try it: B is between A and C with $AB = 4x + 1$, $BC = x + 4$, $AC = 35$. Find x.

check: $\boldsymbol{x = 6}$

Skill 2 — Writing and testing a converse

"If P then Q" $\longrightarrow$ converse "if Q then P". Swap the two halves; change nothing else.

A converse must be judged on its own. A true statement can have a false converse (linear pair $\Rightarrow$ supplementary is true; supplementary $\Rightarrow$ linear pair is false). To disprove one, give a case meeting the new hypothesis that breaks the new conclusion.

Converses do real work: the forward theorem tells you what parallel lines *give*; the converse is what lets you *conclude* lines are parallel.

Example: Corresponding angles measure $(3x + 10)^\circ$ and $(5x - 20)^\circ$. Find x so the lines are parallel.

Step 1. To *conclude* parallel lines I need the converse of the Corresponding Angles Postulate, whose hypothesis is that the two corresponding angles are congruent — so I set the measures equal.

Step 2. $3x + 10 = 5x - 20$, so $30 = 2x$.

$\Rightarrow$ $\boldsymbol{x = 15}$

Try it: Corresponding angles measure $(2x + 25)^\circ$ and $(4x - 15)^\circ$. Find x so the lines are parallel.

check: $\boldsymbol{x = 20}$

Skill 3 — When a biconditional is allowed

"P if and only if Q" asserts **both** "if P then Q" and "if Q then P". Write one only when you have checked both directions — which is why *definitions* are biconditional and most theorems are not.

Reading it in a problem: a biconditional may be used in either direction, so it both supplies a fact and licenses a conclusion.

Example: An angle measures $(8x - 6)^\circ$. Find x that makes it a right angle.
Step 1. "Right angle if and only if the measure is 90°" runs backwards as well as forwards, so I may set the given measure equal to 90 and conclude the angle is right.
Step 2. $8x - 6 = 90$, so $8x = 96$.
$\Rightarrow$ $\boldsymbol{x = 12}$

Try it: An angle measures $(5x + 15)^\circ$. Find x that makes it a right angle.

check: $\boldsymbol{x = 15}$

(!) Watch out: a true statement does not license its converse. Before writing "if and only if", test the backwards direction — one counterexample kills it.